

Back propagation Algo (Regression):

epoch = 5 (You want to enter any epoch).

for i in range (epoch):

for j in range (x.shape [0]):

$\frac{\partial L}{\partial w_{11}} = -2(y - \hat{y}) w_{11}^2 x_{i1}$ → select 1 row (random).

$\frac{\partial L}{\partial w_{11}}$ → predict (using F.B)

→ calculate loss (using loss function - mse)

$\frac{\partial L}{\partial w_{12}} = -2(y - \hat{y}) w_{12}^2 x_{i1}$ → update weights & bias using GD

$\frac{\partial L}{\partial w_{12}}$ $w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

$\frac{\partial L}{\partial w_{11}^2} = -2(y - \hat{y}) 0_{11}$ → calculate avg dw for epoch.

$\frac{\partial L}{\partial w_{11}^2}$

$\frac{\partial L}{\partial w_{21}} = -2(y - \hat{y}) 0_{12}$

$\frac{\partial L}{\partial w_{21}}$

$\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y})$

$\frac{\partial L}{\partial b_{11}}$

Back propagation Algo Classification

eg: spam placement.

Activation function is sigmoid $\frac{1}{1 + e^{-x}}$

$\frac{\partial L}{\partial w_{11}} = -2(y - \hat{y}) w_{11}^2 x_{i1}$

$\frac{\partial L}{\partial w_{11}}$

epoch = 5

for ii in range (epochs):

for j in range (n.shape [0]):

— select 1 row

— calculate F.B.P

— calculate loss

— update w & b

$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w}$

— calculate average loss.

Derivative of sigmoid
 $= \sigma(z) (1 - \sigma(z))$

$$\frac{\partial L}{\partial \hat{y}} = \frac{-(y - \hat{y})}{\hat{y}(1 - \hat{y})}$$

$$\frac{\partial \hat{y}}{\partial z} = \frac{\partial}{\partial z} (\sigma(z))$$

$$= \sigma(z) (1 - \sigma(z)) \quad \because \hat{y} = \sigma(z)$$

$$\frac{\partial \hat{y}}{\partial z} = \hat{y} (1 - \hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} = \hat{y} - y \quad \text{used in (a)} \\ \text{or } -(y - \hat{y}) \quad \text{(b) \& (c)}$$

$$\frac{\partial z}{\partial w_{11}^2} = \frac{\partial}{\partial w_{11}^2} [\omega_{11}^2 O_{11} + \omega_{21}^2 O_{12} + b_{21}]$$

$$\frac{\partial z}{\partial w_{11}^2} = O_{11}$$

As

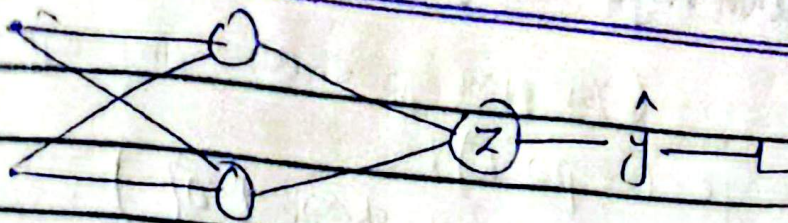
$$\frac{\partial z}{\partial w_{21}^2} = O_{12}$$

As

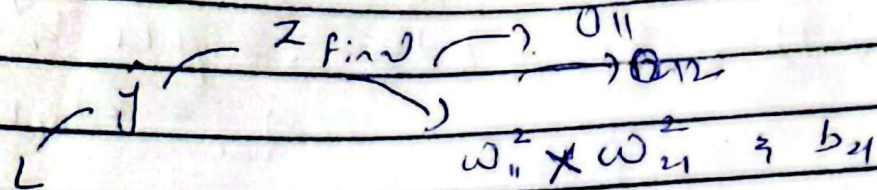
$$\frac{\partial z}{\partial b_{21}} = 1$$

put in (a), (b) \& (c)

$$\frac{\partial L}{\partial w_{11}^2} = -(y - \hat{y}) O_{11}$$



$$z_{final} = \omega_{11}^2 o_{11} + \omega_{21}^2 o_{12} + b_{21}$$



$$\frac{\partial L}{\partial \omega_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial \omega_{11}^2} \quad \text{--- (a)}$$

$$\frac{\partial L}{\partial \omega_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial \omega_{21}^2} \quad \text{--- (b)}$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial b_{21}} \quad \text{--- (c)}$$

$$\frac{\partial L}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} (-y \log(\hat{y}) - (1-y) \log(1-\hat{y}))$$

$$= \frac{-y}{\hat{y}} - \frac{1-y}{1-\hat{y}} (-1)$$

$$= \frac{-y(1-\hat{y}) + (1-y)\hat{y}}{\hat{y}(1-\hat{y})}$$

$$= \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$\frac{\partial L}{\partial w_{11}} = -(y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12}) x_{11}$$

As for $\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial b_{12}}$

$$\frac{\partial L}{\partial w_{12}} = -(y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12}) x_{12}$$

$$\frac{\partial L}{\partial b_{12}} = -(y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12})$$

THE WHY

Back propagation Algorithm.

epoch = 5

#epoch x row

for i in range(epoch):

for j in range(x.shape[0]):

- select randomly 1 point/row
- Predict (using forward propagation).
- Calculate loss (using L-Function).
- Update weights & bias using

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

$$b_{\text{new}} = b_{\text{old}} - \eta \frac{\partial L}{\partial b}$$

→ Calculate avg loss of function

$$\frac{\partial L}{\partial w_{21}^2} = -(y - \hat{y}) O_{12}$$

$$\frac{\partial L}{\partial b_{21}} = -(y - \hat{y})$$

$$\bullet \frac{\partial L}{\partial w'_{11}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_{final}} \times \frac{\partial z_{final}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial z_{prev}} \times \frac{\partial z_{prev}}{\partial w'_{11}} \quad \text{--- (1)}$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_{final}} = -(y - \hat{y})$$

$$\frac{\partial z_{final}}{\partial O_{11}} = \frac{\partial}{\partial O_{11}} [w_{11}^2 O_{11} + w_{21}^2 O_{12} + b_{21}]$$

$$= w_{11}^2$$

$$\frac{\partial O_{11}}{\partial z_{prev}} = \frac{\partial}{\partial z_{prev}} (\sigma(z_{prev}))$$

$$= \sigma'(z_{prev}) = (1 - \sigma(z_{prev}))$$

$$\frac{\partial O_{11}}{\partial z_{prev}} = O_{11} (1 - O_{11})$$

$$\frac{\partial z_{prev}}{\partial w'_{11}} = \frac{\partial}{\partial w'_{11}} [w'_{11} x_{i1} + w'_{21} x_{i2} + b_{11}]$$

$$= x_{i1}$$

$$P_{11} \sim (1)$$

$$\frac{\partial L}{\partial \omega_{11}} = -(\hat{y} - y) \omega_{11}^2 x_{i1} o_{11} (1 - o_{11})$$

As for $\frac{\partial L}{\partial \omega_{21}} \& \frac{\partial L}{\partial b_{11}}$

$$\frac{\partial L}{\partial \omega_{21}} = -(\hat{y} - y) \omega_{11}^2 o_{11} (1 - o_{11}) x_{i2}$$

$$\frac{\partial L}{\partial b_{11}} = -(\hat{y} - y) \omega_{11}^2 o_{11} (1 - o_{11})$$

• $\frac{\partial L}{\partial \omega'_{12}}, \frac{\partial L}{\partial \omega'_{22}}, \frac{\partial L}{\partial b_{12}}$

$$\frac{\partial L}{\partial \omega'_{12}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_{final}} \times \frac{\partial z_{final}}{\partial o_{12}} \times \frac{\partial o_{12}}{\partial z_{prev}} \times \frac{\partial z_{prev}}{\partial \omega'_{12}}$$

$$\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_{final}} = -(\hat{y} - y)$$

$$\frac{\partial z_f}{\partial o_{12}} = \frac{\partial}{\partial o_{12}} \left[\omega_{11}^2 o_{11} + \omega_{21}^2 o_{12} + b_{21} \right]$$

$$\frac{\partial z_f}{\partial o_{12}} = \omega_{21}^2$$

$$\frac{\partial z_p}{\partial \omega'_{12}} = \frac{\partial}{\partial \omega'_{12}} \left(\omega'_{12} x_{i1} + \omega'_{22} x_{i2} + b_{12} \right)$$

$$\frac{\partial o_{12}}{\partial z_p} = \frac{\partial}{\partial z_p} (\sigma(z_p))$$

$$= \sigma(z_p) (1 - \sigma(z_p))$$

$$= o_{12} (1 - o_{12})$$

$$\frac{\partial z_p}{\partial \omega'_{12}} = x_{i1}$$